NovaWireless KPI Drift Observatory

Goodhart's Law in Action

"When a measure becomes a target, it ceases to be a good measure." — Goodhart, 1975

Addendum to the NovaWireless Research Series
March 2026

Dataset: 60,000 synthetic calls • Corrected ledger • 12-month observation window

Abstract

This addendum presents the evidence for Goodhart's Law operating within the NovaWireless call center environment. Across 60,000 synthetic calls spanning 12 months, we observe a persistent 20.2 percentage-point gap between proxy First Contact Resolution (58.2%) and true durable resolution (38.0%). The proxy metric reports a green dashboard while customer trust erodes, repeat contacts accumulate at approximately 20% within 30 days (and an additional 24% within 31–60 days), and representatives learn that gaming the proxy is more reliably rewarded than genuinely solving customer problems.

Six figures document the complete Goodhart mechanism: the dashboard lie, the scenario contradiction, the KPI-trust divergence, the bandaid credit economy, gaming propensity drift, and a summary panel synthesizing all signals.

1. The Green Dashboard Lie

Figure 1 establishes the foundational observation. Proxy FCR reports 58.2%—a number that would appear acceptable on any standard call center dashboard. True FCR—measured as durable resolution with no repeat contact, escalation, or churn within the observation window—is 38.0%. The gap is 20.2 percentage points. Over the 12-month observation window, this gap remains stable: the proxy hovers near 90% while true resolution remains below 50%. The system has reached a steady-state lie.

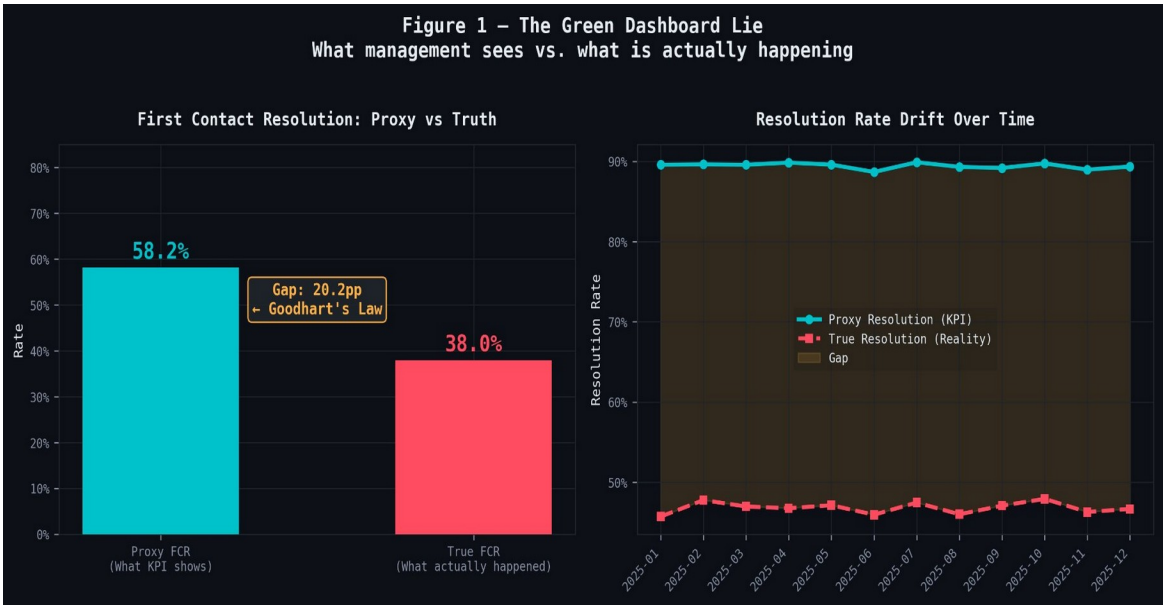

Figure 1 — The Green Dashboard Lie. Proxy FCR (58.2%) vs. True FCR (38.0%). The 20.2pp gap is Goodhart's Law in action.

2. The Scenario Contradiction

Figure 2 decomposes the gap by call scenario. Standard clean calls show proxy and true resolution tracking closely. But fraud scenarios (store promo, care promo, line add, HIC exchange) and gamed_metric calls show massive divergence—proxy resolution rates near 100% while true resolution rates fall below 20%. The gamed_metric scenario is the most

extreme: proxy shows nearly perfect resolution while true resolution is approximately 14%. These scenarios maintain a green dashboard precisely because the gaming mechanism targets the proxy measurement.

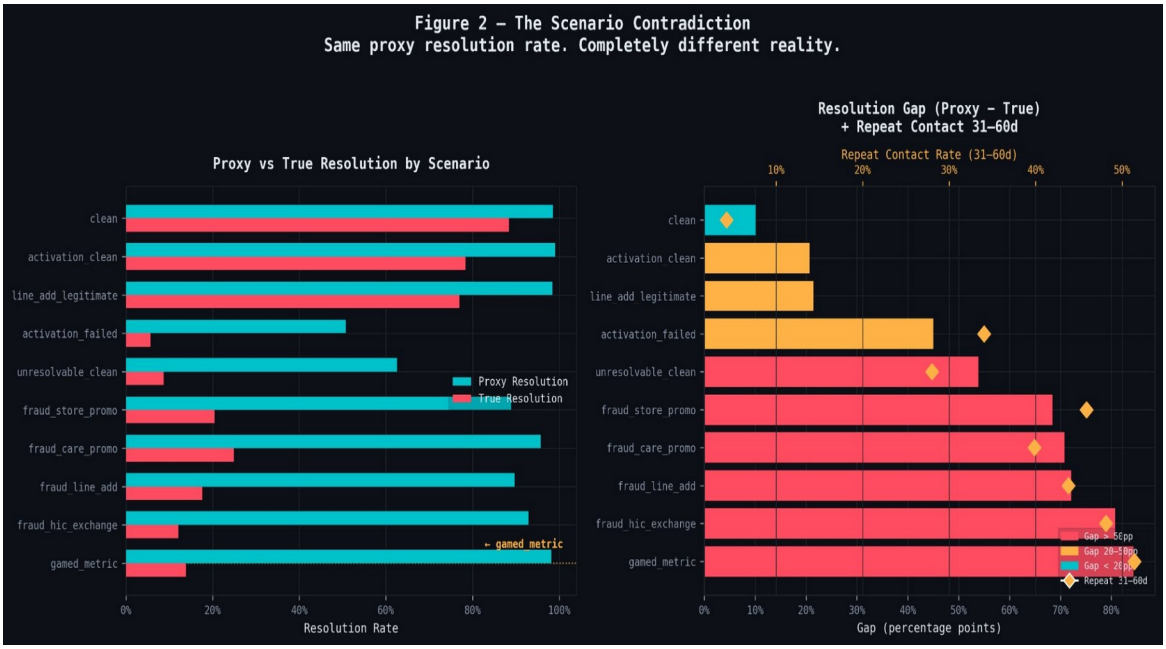

Figure 2 — The Scenario Contradiction. Same proxy resolution rate, completely different reality across scenarios.

3. KPI Lift vs. Reality Decay

Figure 3 presents the four-panel temporal view. Top-left: proxy FCR and customer trust trend in opposite directions—as the metric climbs, trust decays. Top-right: AHT rises over the observation period, suggesting calls are becoming more complex or contentious even as proxies report success. Bottom-left: repeat contact rates remain stubbornly persistent at approximately 20% (30-day) and 24% (31–60 day), representing hidden cost that appears on no standard KPI dashboard. Bottom-right: representative gaming propensity and burnout both fluctuate but show no corrective pressure from the KPI system.

Figure 3 — KPI Lift vs. Reality Decay. Green metrics climb; customer trust erodes; repeat contacts persist; reps adapt.

4. The Bandaid Economy

Figure 4 exposes the credit mechanism. Of all calls, 44.2% receive no credit. Among those that do, service credits (15.9%), dispute credits (14.5%), and courtesy credits (12.8%) dominate. Bandaid credits—unauthorized hush money applied to silence a complaint without fixing the underlying issue—account for 8.7%. The center panel shows the bandaid rate fluctuating month-over-month alongside the bandaid failure rate (customers who received a bandaid credit and called back within 31–60 days). The right panel shows a positive correlation (r = 0.174) between a representative's bandaid rate and their repeat contact rate, colored by gaming propensity—higher gamers issue more bandaids and generate more repeat contacts.

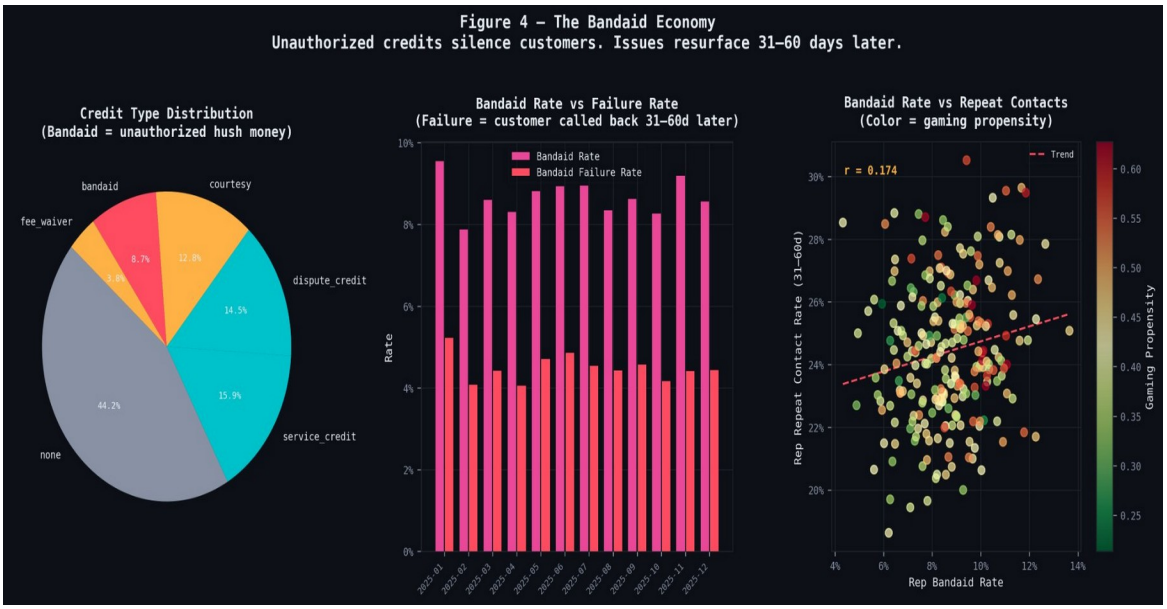

Figure 4 — The Bandaid Economy. Unauthorized credits silence customers temporarily; issues resurface 31–60 days later.

5. Gaming Propensity Drift

Figure 5 maps the gaming landscape across all 250 representatives. The left panel shows the distribution of average gaming propensity: 196 reps fall below the 0.5 threshold, while 54 reps operate at or above it. The right panel reveals a strong positive correlation (r = 0.763) between gaming propensity and the resolution gap (proxy minus true FCR), with color indicating true FCR—greener dots (higher true FCR) cluster at lower gaming propensity, while redder dots (lower true FCR) dominate the high-gaming region. Representatives with higher gaming propensity systematically produce larger proxy-truth gaps.

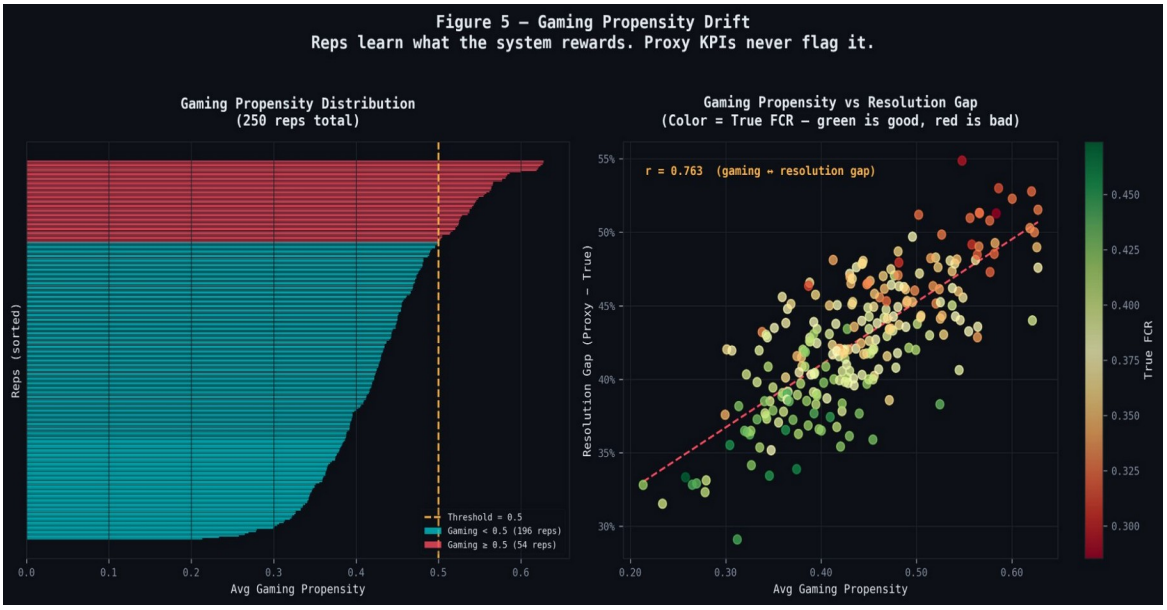

Figure 5 — Gaming Propensity Drift. More gaming = bigger KPI lie. Proxy KPIs never flag it.

6. Goodhart Summary Panel

Figure 6 synthesizes all findings into a single executive view. Panel A: the 20.2pp FCR gap. Panel B: the five worst-gap scenarios. Panel C: proxy FCR climbing while customer trust decays. Panel D: persistent repeat contacts hidden from dashboards. Panel E: bandaid credits correlating with repeat contacts. Panel F: gaming propensity driving the resolution gap (r = 0.763). Together, these six views constitute a complete evidence chain for Goodhart's Law operating at system scale in a call center environment.

Figure 6 — Goodhart Summary Panel. Six views of the same dysfunction: the metric became the target and ceased to measure reality.

7. Conclusion

The NovaWireless KPI Drift Observatory confirms that Goodhart's Law is not a theoretical curiosity—it is an operational reality observable at every level of the system. The proxy FCR metric has been so thoroughly gamed that it provides zero informational value about actual customer outcomes. Representatives who game more produce larger gaps. Bandaid credits create temporary silence that resurfaces as repeat contacts. Customer trust decays while the dashboard stays green.

This addendum establishes the empirical foundation for the System Integrity Index (SII) proposed in the companion trust signal health assessment. Without a governance mechanism that monitors the gap between proxy and truth—rather than the proxy alone—any optimization effort (human or AI) will inevitably drift toward Goodhart failure.

Appendix: Dataset Specification

60,000 synthetic call records generated via run_lab.py. Corrected master account ledger (~8% IMEI mismatch). billing_agreement_type fix applied. 250 representatives, 5,000 customers, 12-month observation window (January–December 2025).